Shren Patel

Staff AI Engineer — Stateful Multi-Agent Orchestration

patel.shrenil@gmail.com · +1 613-551-9989 · Canada remote · NA-friendly · EST/PST
linkedin.com/in/shrenp · github.com/blopit · shrenil.com

SUMMARY

10+ years shipping software, 3+ on production AI agents. Built CentrOS — stateful multi-agent runtime: LangGraph state graphs, Temporal-style durability, Pydantic tool contracts, LangFuse traces, hybrid RAG, HITL approvals; 3,980+ Vitest tests on main, 0 silent run loss across 6 months. Exact architecture this role calls for.

EXPERIENCE

Founder CentrOS — AI Agent Orchestration Platform *Feb 2026 – Present*
Remote · Canada • TypeScript · Hono · PostgreSQL · Drizzle · Vitest · LLM tool-calling

- Built CentrOS production multi-agent orchestration runtime: typed tool contracts, eval gates, LangFuse traces, HITL approval interrupts — collapsed 4 vendor primitives into 1 governed layer.
- Designed agent runtime on LangGraph state graphs + Temporal-style durability: durable execution, checkpointing, crash recovery, retries, versioned rollouts — 0 silent run loss across 6 months.
- Built eval harness + LangFuse trace pipeline: golden datasets, task-level scorers, regression suites, prompt versioning, failure taxonomy — caught 22+ silent regressions before release.
- Implemented hybrid RAG pipelines on pgvector + OpenSearch: query rewriting, source ranking, chunk hygiene, freshness boundaries, selective context packing — cut prompt token use 40% with grounding intact.
- Designed HITL interrupt model: confidence-gated handoff, approval checkpoints before irreversible actions, reversible operations, audit trails, escalation policy — 100% of high-risk tool calls human-reviewed.

Independent Engineer / Builder Software, AI automation & creative IP *Sep 2022 – Jan 2026*
GTA · Remote • Python · TypeScript · Node · LLM tool-calling · MCP · multi-agent orchestration

- Built LLM workflow automation since ChatGPT launch (Nov 2022): tool-calling/API automations from 2023; LangGraph multi-agent patterns from 2024; MCP integrations from late 2024 — 3+ years compounding.

Senior Frontend Developer Faire *Jun 2021 – Aug 2022*
Remote · Canada • React · TypeScript · MobX · Jest · Storybook · A/B testing

- Drove site-wide perf gains via A/B testing + staged rollouts; same guardrail discipline now applied to agent versioning, shadow traffic, and canary releases at CentrOS — 5+ wins shipped.

iOS Mobile & Web Developer Coursera *Jan 2017 – Apr 2017*
Mountain View, CA • React · Python · iOS SDK · Swift · NLP/search

- Built Python search tooling lifting course content discovery 60% (pre-LLM NLP era) — same retrieval problem CentrOS now solves with embeddings, hybrid RAG, and pgvector.

SKILLS

AI / Agents:

- LangGraph state graphs · Temporal-style durability · Pydantic tool contracts · LangFuse tracing
- Multi-agent orchestration · task decomposition · router/executor/verifier patterns
- RAG · retrieval pipelines · embeddings · context optimization · prompt management
- Human-in-the-loop approvals · audit trails · governance · eval-gated rollout
- Tool & function calling · structured outputs · MCP servers

Languages:

- Python 3 (primary) · TypeScript · Node.js

Cloud & Infra:

- GCP · AWS · Kubernetes · Docker · serverless · CI/CD
- Vector DBs: pgvector · OpenSearch · FAISS

EDUCATION

University of Waterloo Bachelor of Software Engineering (BSE) — Computer Software Engineering *2012 – 2018*
Waterloo, ON